

Simulation Study: Three-Time-Point SNTTEM

Data-Generating Mechanism and Results by Scenario

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1 Data-Generating Mechanism

1.1 Common Temporal Structure

We simulate longitudinal data for n individuals over three time points $t \in \{0, 1, 2\}$. At each time point we observe a time-varying covariate L_t , an eligibility indicator $I_t \in \{0, 1\}$, and a binary treatment $A_t \in \{0, 1\}$. The temporal ordering is

$$I_0, L_0, A_0 \longrightarrow I_1, L_1, A_1 \longrightarrow I_2, L_2, A_2 \longrightarrow Y.$$

The DGP is governed by five scalar parameters:

- $\psi_{03}, \psi_{13}, \psi_{23}$: structural blip parameters encoding the causal effect of treatment at each time point;
- $p_I \in (0, 1)$: marginal probability of eligibility at times 1 and 2;
- $p_h \in (0, 1)$: probability that treatment is *held* (i.e. unchanged) from one period to the next.

1.2 Time $t = 0$

Every individual is eligible at baseline ($I_0 = 1$).

$$L_0 \sim \text{Uniform}(0, 0.5), \quad A_0 \mid L_0 \sim \text{Bernoulli}(\text{expit}(-1 + L_0)),$$

where $\text{expit}(x) = e^x / (1 + e^x)$.

1.3 Times $t = 1$ and $t = 2$

The covariate L_t is a truncated version of a latent normal draw. We use the clipping operator $\text{clip} : \mathbb{R} \rightarrow [a, b]$, defined for bounds $a < b$ as

$$\text{clip}(x; a, b) = \max(a, \min(b, x)) = \begin{cases} a & \text{if } x < a, \\ x & \text{if } a \leq x \leq b, \\ b & \text{if } x > b. \end{cases} \quad (1)$$

This operation hard-limits the support of L_t to the interval $[a, b]$, replacing any draw below a with a and any draw above b with b , while leaving values already within $[a, b]$ unchanged.

For each $t \in \{1, 2\}$:

$$\begin{aligned} I_t &\sim \text{Bernoulli}(p_I), \\ L_t^* \mid A_{t-1} &\sim \mathcal{N}(-0.5 A_{t-1}, 1), \\ L_t &= \text{clip}(L_t^*; -1.5, -0.5), \\ A_t \mid A_{t-1} &= \begin{cases} A_{t-1} & \text{with probability } p_h, \\ 1 - A_{t-1} & \text{with probability } 1 - p_h. \end{cases} \end{aligned}$$

The clipping bounds $[-1.5, -0.5]$ ensure that L_t has bounded, strictly negative support regardless of A_{t-1} . Concretely: without treatment ($A_{t-1} = 0$) the latent draw is $L_t^* \sim \mathcal{N}(0, 1)$, and with treatment ($A_{t-1} = 1$) it is $L_t^* \sim \mathcal{N}(-0.5, 1)$; in both cases the clip truncates the distribution to $[-1.5, -0.5]$, keeping the covariate in a regime where the blip functions γ_{13} and γ_{23} (which depend on $1 + L_t \in [0.5, 1.5]$) remain positive and bounded away from zero.

1.4 Structural Blip Functions

The structural blip functions represent the causal effect on the outcome of incrementing treatment by one unit at time t , conditional on the covariate history:

$$\gamma_{03}(L_0; \psi_{03}) = \psi_{03} L_0, \quad (2)$$

$$\gamma_{13}(L_1; \psi_{13}) = \psi_{13} (1 + L_1), \quad (3)$$

$$\gamma_{23}(L_2; \psi_{23}) = \psi_{23} (1 + L_2). \quad (4)$$

1.5 Binary Outcome DGP

Used in Scenarios S1 (null) and S2 (non-null). The outcome follows a log-linear model $Y \sim \text{Bernoulli}(\exp(\eta))$, where

$$\begin{aligned} \eta = & -1.5 + \psi_{03} L_0 A_0 \\ & + I_1 \psi_{13} (1 + L_1) (A_1 - A_0) + (1 - I_1) \psi_{13} \frac{L_1}{3} (A_1 - A_0) \\ & + I_2 \psi_{23} (1 + L_2) (A_2 - A_1) + (1 - I_2) \psi_{23} \frac{L_2}{3} (A_2 - A_1). \end{aligned} \quad (5)$$

For eligible individuals ($I_t = 1$) the blip is $\gamma_t(L_t)$; for ineligible individuals ($I_t = 0$) it is attenuated to $\gamma_t(L_t) \cdot L_t / [3(1 + L_t)]$.

1.6 Continuous Outcome DGP

Used in Scenarios S3–S5 ($\psi = 1$, varying noise). The outcome follows a Gaussian model $Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$, where

$$\mu_Y = -1.5 + \psi_{03} L_0 A_0 + \psi_{13} (1 + L_1) (A_1 - A_0) + \psi_{23} (1 + L_2) (A_2 - A_1), \quad (6)$$

$$\sigma_Y = \sigma [A_0 + (A_1 - A_0)^2 + (A_2 - A_1)^2] + 1. \quad (7)$$

Unlike the binary DGP, the mean outcome (6) contains no eligibility interaction: the blip $\gamma_t(L_t)$ applies equally to all individuals. The heteroscedasticity parameter $\sigma \geq 0$ scales noise with treatment activity; $\sigma = 0$ gives a deterministic outcome.

2 Nuisance Models

All estimators rely on the following nuisance functions, estimated via two-fold cross-fitting in each Monte Carlo iteration:

- Propensity scores $\pi_t = P(A_t = 1 \mid \bar{L}_t, A_{t-1})$, $t \in \{0, 1, 2\}$.
- Outcome regressions μ_{t3} constructed sequentially from the SNTTEM backward induction, using doubly-robust (DR) pseudo-outcomes at each step.

Within each fold, each nuisance function is estimated by an equal-weight ensemble of:

1. Random Forest (200 trees);
2. Gradient Boosting Machine (200 trees, max depth 4, learning rate 0.1);
3. Degree-3 polynomial Logistic/Linear Regression.

3 Estimators

Three estimators of $\boldsymbol{\psi} = (\psi_{03}, \psi_{13}, \psi_{23})$ are compared. All solve a system of three estimating equations via M-estimation (root-finding with `scipy.optimize.fsolve`).

Three-step estimator (TS). Conventional sequential regression approach. Uses eligibility-based SNTTEM blip-down with multiplicative (binary) or linear (continuous) transformations.

Simple g-estimator (SG). Doubly-robust SNTTEM estimator. Uses eligibility-modified weights $w_t^I = w_t^{1-I_t}$ in all three estimating equations, exploiting the eligibility indicator to identify blip parameters only from eligible subjects while borrowing efficiency from ineligible subjects through the outcome regression.

GMM estimator. Optimal inverse-covariance combination of TS and SG, computed via 200 bootstrap resamples of the nuisance-fitted data per iteration.

4 Simulation Design

All scenarios cross $p_h \in \{0.4, 0.6, 0.8\}$ with $p_I \in \{0.25, 0.50, 0.75, 1.00\}$, giving 12 cells per scenario. Each cell uses 100 Monte Carlo iterations (seed 3411). Table 1 summarises the five scenarios.

Table 1: Summary of simulation scenarios.

Scenario	Outcome	True $\boldsymbol{\psi}$	σ	n	Estimating eqs.
S1	Binary	(0, 0, 0)	—	6000	Multiplicative blip-down
S2	Binary	(1, 1, 1)	—	6000	Multiplicative blip-down
S3	Continuous	(1, 1, 1)	0	4000	Linear blip-down
S4	Continuous	(1, 1, 1)	10	4000	Linear blip-down
S5	Continuous	(1, 1, 1)	20	4000	Linear blip-down

Under the null ($\psi = \mathbf{0}$) the multiplicative and linear estimating equations are algebraically equivalent (since $e^{-\gamma_t \cdot 0} = 1$ and $-\gamma_t \cdot 0 = 0$), so S1 applies to both binary and continuous estimators simultaneously.

5 Results

Tables 2–13 report empirical *bias* ($\hat{\psi} - \psi_{\text{true}}$) and Monte Carlo *variance* over 100 iterations for each method, scenario, and parameter. Results are grouped by p_h within each table.

5.1 Scenario S1: Null ($\psi_{03} = \psi_{13} = \psi_{23} = 0$, Binary DGP, $n = 6000$)

Table 2: S1 — Empirical bias and variance for $\hat{\psi}_{03}$. True value: $\psi_{03} = 0$.

p_h	p_I	Three-step		Simple g-est.		GMM	
		Bias	Var	Bias	Var	Bias	Var
0.4	0.25	0.0149	0.5701	0.0158	0.4560	−0.1452	0.4505
	0.50	0.0144	0.5681	0.0260	0.3945	−0.1046	0.3706
	0.75	0.0139	0.5675	0.0135	0.3220	−0.0865	0.2962
	1.00	0.0145	0.5687	0.0069	0.2397	−0.0583	0.2344
0.6	0.25	0.0122	0.1811	0.0132	0.1516	−0.0486	0.1463
	0.50	0.0130	0.1811	0.0143	0.1398	−0.0432	0.1362
	0.75	0.0127	0.1812	0.0177	0.1387	−0.0190	0.1365
	1.00	0.0125	0.1812	0.0159	0.1164	−0.0191	0.1083
0.8	0.25	0.0077	0.0948	0.0070	0.0855	−0.0205	0.0841
	0.50	0.0080	0.0947	0.0082	0.0826	−0.0180	0.0791
	0.75	0.0078	0.0946	0.0097	0.0848	−0.0151	0.0849
	1.00	0.0076	0.0948	0.0063	0.0763	−0.0162	0.0740

5.2 Scenario S2: Binary Outcome, Non-null ($\psi_{03} = \psi_{13} = \psi_{23} = 1$, $n = 6000$)

5.3 Scenario S3: Continuous Outcome, $\sigma = 0$ ($\psi = 1$, $n = 4000$)

When $\sigma = 0$, the outcome is deterministic given the treatment and covariate history (variance of Y equals 1 regardless of treatment, since $\sigma_Y = 0 \cdot (\dots) + 1 = 1$; in practice $\sigma = 0$ collapses the noise to its floor). All three estimators recover the true parameters with negligible Monte Carlo bias ($|\text{Bias}| < 5 \times 10^{-4}$) and near-zero variance ($< 10^{-6}$) across all 12 (p_h, p_I) cells. No numerical table is presented, but the corresponding plots are shown in Figures 5 and 6 to confirm the near-zero performance visually.

Table 3: S1 — Empirical bias and variance for $\hat{\psi}_{13}$. True value: $\psi_{13} = 0$.

p_h	p_I	Three-step		Simple g-est.		GMM	
		Bias	Var	Bias	Var	Bias	Var
0.4	0.25	0.0598	0.4521	0.0512	0.3721	-0.0386	0.3879
	0.50	0.0042	0.2284	-0.0035	0.1712	-0.0678	0.1696
	0.75	0.0007	0.1437	-0.0073	0.1195	-0.0550	0.1164
	1.00	0.0001	0.1060	-0.0107	0.0948	-0.0489	0.0924
0.6	0.25	0.0246	0.2917	0.0240	0.2775	-0.0376	0.2771
	0.50	0.0128	0.1460	0.0141	0.1276	-0.0286	0.1245
	0.75	0.0220	0.0859	0.0183	0.0748	-0.0056	0.0735
	1.00	0.0127	0.0711	0.0116	0.0645	-0.0090	0.0611
0.8	0.25	0.0488	0.3419	0.0482	0.3224	-0.0085	0.3266
	0.50	0.0345	0.1432	0.0359	0.1331	0.0080	0.1282
	0.75	0.0096	0.0775	0.0074	0.0708	-0.0200	0.0711
	1.00	0.0095	0.0677	0.0104	0.0610	-0.0084	0.0593

Table 4: S1 — Empirical bias and variance for $\hat{\psi}_{23}$. True value: $\psi_{23} = 0$. TS and SG share the same third estimating equation, so their $\hat{\psi}_{23}$ estimates are identical by construction.

p_h	p_I	Three-step		Simple g-est.		GMM	
		Bias	Var	Bias	Var	Bias	Var
0.4	0.25	-0.0203	0.2057	-0.0203	0.2057	-0.0321	0.1914
	0.50	-0.0195	0.0927	-0.0195	0.0927	-0.0354	0.0916
	0.75	-0.0021	0.0499	-0.0021	0.0499	-0.0200	0.0502
	1.00	-0.0061	0.0406	-0.0061	0.0406	-0.0221	0.0401
0.6	0.25	-0.0235	0.1998	-0.0235	0.1998	-0.0247	0.1862
	0.50	-0.0139	0.0896	-0.0139	0.0896	-0.0265	0.0881
	0.75	0.0037	0.0506	0.0037	0.0506	-0.0061	0.0499
	1.00	-0.0026	0.0416	-0.0026	0.0416	-0.0108	0.0405
0.8	0.25	0.0200	0.2698	0.0200	0.2698	0.0264	0.2494
	0.50	-0.0055	0.1341	-0.0055	0.1341	-0.0119	0.1282
	0.75	-0.0040	0.0757	-0.0040	0.0757	-0.0107	0.0715
	1.00	0.0002	0.0588	0.0002	0.0588	-0.0043	0.0559

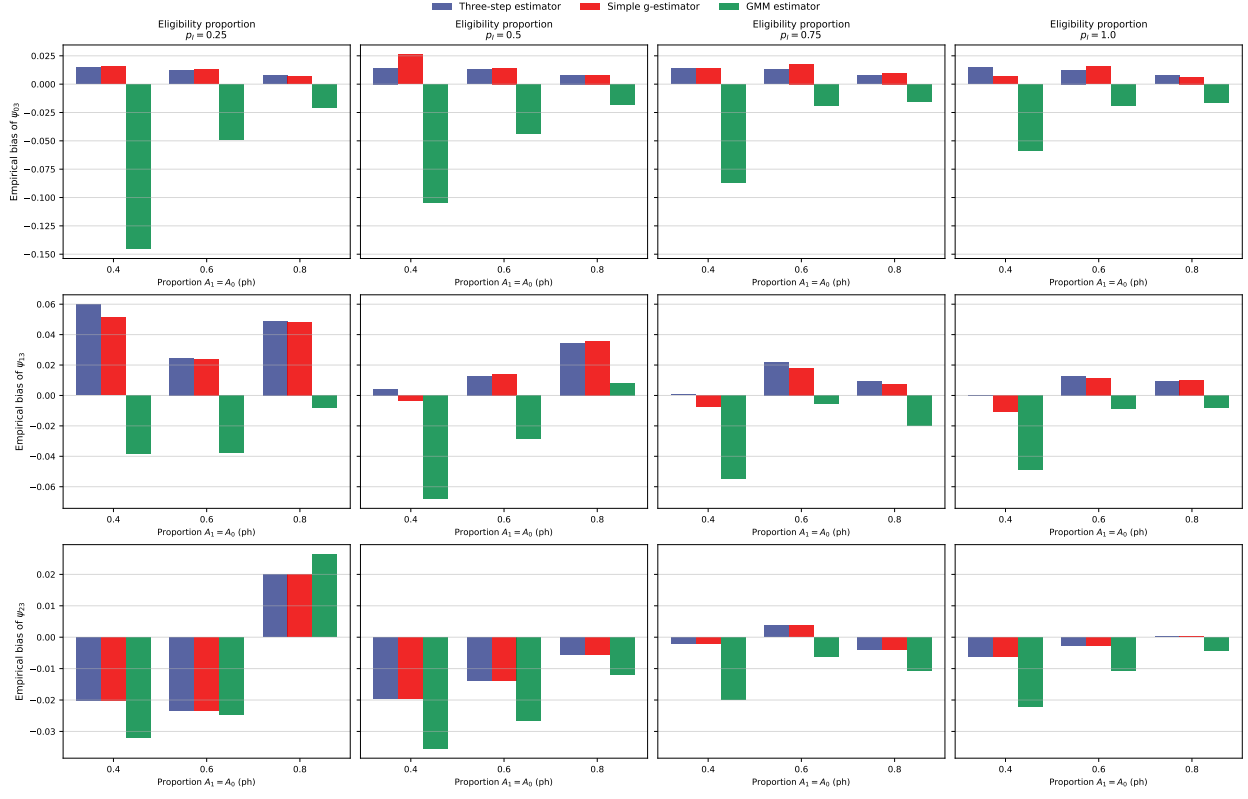


Figure 1: S1 (null, binary DGP) — Empirical bias of $\hat{\psi}_{03}$, $\hat{\psi}_{13}$, and $\hat{\psi}_{23}$ (rows) as a function of p_h (x-axis), stratified by eligibility proportion p_I (columns). Grouped bars show the Three-step estimator (blue), Simple g-estimator (red), and GMM estimator (green). True values $\psi_{03} = \psi_{13} = \psi_{23} = 0$.

Table 5: S2 — Empirical bias and variance for $\hat{\psi}_{03}$. True value: $\psi_{03} = 1$.

p_h	p_I	Three-step		Simple g-est.		GMM	
		Bias	Var	Bias	Var	Bias	Var
0.4	0.25	0.1529	0.6120	0.2427	0.5465	0.0359	0.5626
	0.50	0.1700	0.6085	0.3033	0.4548	0.1040	0.4189
	0.75	0.1586	0.5658	0.2376	0.3231	0.0993	0.3335
	1.00	0.1693	0.5769	0.0935	0.1924	0.0231	0.1974
0.6	0.25	-0.0161	0.2272	0.0762	0.2068	-0.0318	0.2131
	0.50	-0.0166	0.2210	0.1190	0.1885	-0.0016	0.1943
	0.75	-0.0127	0.2214	0.0916	0.1864	0.0278	0.1831
	1.00	-0.0114	0.2240	0.0526	0.1414	0.0037	0.1339
0.8	0.25	-0.0019	0.0973	0.0385	0.0916	-0.0164	0.0920
	0.50	-0.0032	0.0971	0.0568	0.0925	0.0045	0.0934
	0.75	-0.0032	0.0974	0.0487	0.0936	0.0093	0.0942
	1.00	-0.0027	0.0952	0.0282	0.0789	-0.0003	0.0777

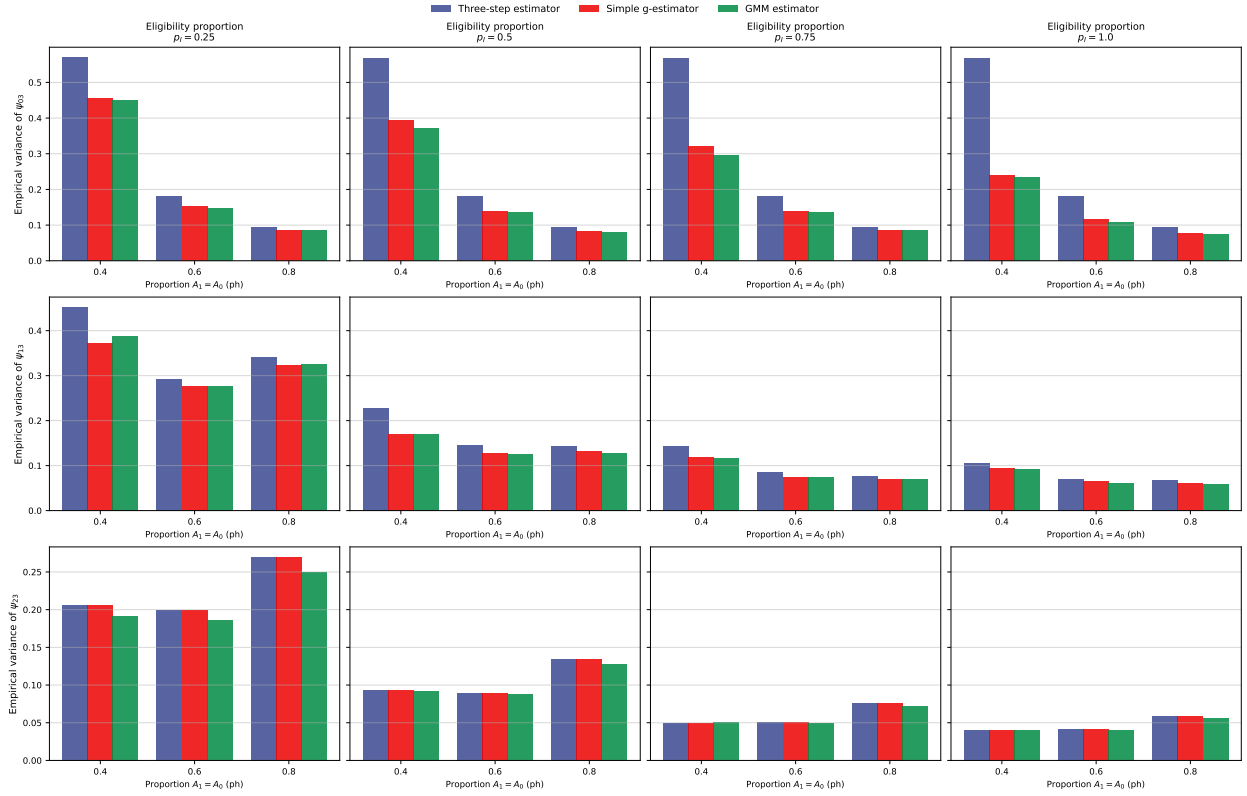


Figure 2: S1 (null, binary DGP) — Empirical variance of $\hat{\psi}_{03}$, $\hat{\psi}_{13}$, and $\hat{\psi}_{23}$ (rows) as a function of p_h , stratified by p_I (columns). Same layout as Figure 1.

Table 6: S2 — Empirical bias and variance for $\hat{\psi}_{13}$. True value: $\psi_{13} = 1$.

p_h	p_I	Three-step		Simple g-est.		GMM	
		Bias	Var	Bias	Var	Bias	Var
0.4	0.25	0.2780	0.4177	0.2461	0.3830	0.2161	0.4209
	0.50	0.2351	0.1832	0.2020	0.1374	0.1483	0.1500
	0.75	0.1510	0.1069	0.1127	0.0983	0.0639	0.0999
	1.00	0.0566	0.0767	0.0396	0.0696	0.0015	0.0747
0.6	0.25	0.2528	0.3428	0.2420	0.3081	0.1763	0.2951
	0.50	0.1896	0.1900	0.1631	0.1677	0.1038	0.1638
	0.75	0.0936	0.0788	0.0762	0.0728	0.0492	0.0684
	1.00	0.0269	0.0782	0.0202	0.0727	0.0010	0.0685
0.8	0.25	0.2077	0.3600	0.2117	0.3316	0.1110	0.3016
	0.50	0.1728	0.1373	0.1610	0.1453	0.0932	0.1336
	0.75	0.1095	0.0945	0.0852	0.0875	0.0422	0.0804
	1.00	0.0365	0.0626	0.0222	0.0592	-0.0018	0.0560

Table 7: S2 — Empirical bias and variance for $\hat{\psi}_{23}$. True value: $\psi_{23} = 1$. TS and SG are identical for $\hat{\psi}_{23}$ by construction.

p_h	p_I	Three-step		Simple g-est.		GMM	
		Bias	Var	Bias	Var	Bias	Var
0.4	0.25	-0.0241	0.1636	-0.0241	0.1636	-0.0335	0.1705
	0.50	0.0115	0.0583	0.0115	0.0583	-0.0044	0.0603
	0.75	0.0148	0.0465	0.0148	0.0465	-0.0049	0.0471
	1.00	0.0024	0.0333	0.0024	0.0333	-0.0140	0.0331
0.6	0.25	0.0223	0.1078	0.0223	0.1078	0.0078	0.1043
	0.50	0.0108	0.0606	0.0108	0.0606	-0.0155	0.0558
	0.75	-0.0012	0.0511	-0.0012	0.0511	-0.0136	0.0511
	1.00	0.0058	0.0414	0.0058	0.0414	-0.0007	0.0398
0.8	0.25	0.0313	0.2726	0.0313	0.2726	-0.0090	0.2136
	0.50	0.0409	0.1090	0.0409	0.1090	0.0297	0.1070
	0.75	0.0206	0.0666	0.0206	0.0666	0.0157	0.0666
	1.00	0.0183	0.0563	0.0183	0.0563	0.0143	0.0539

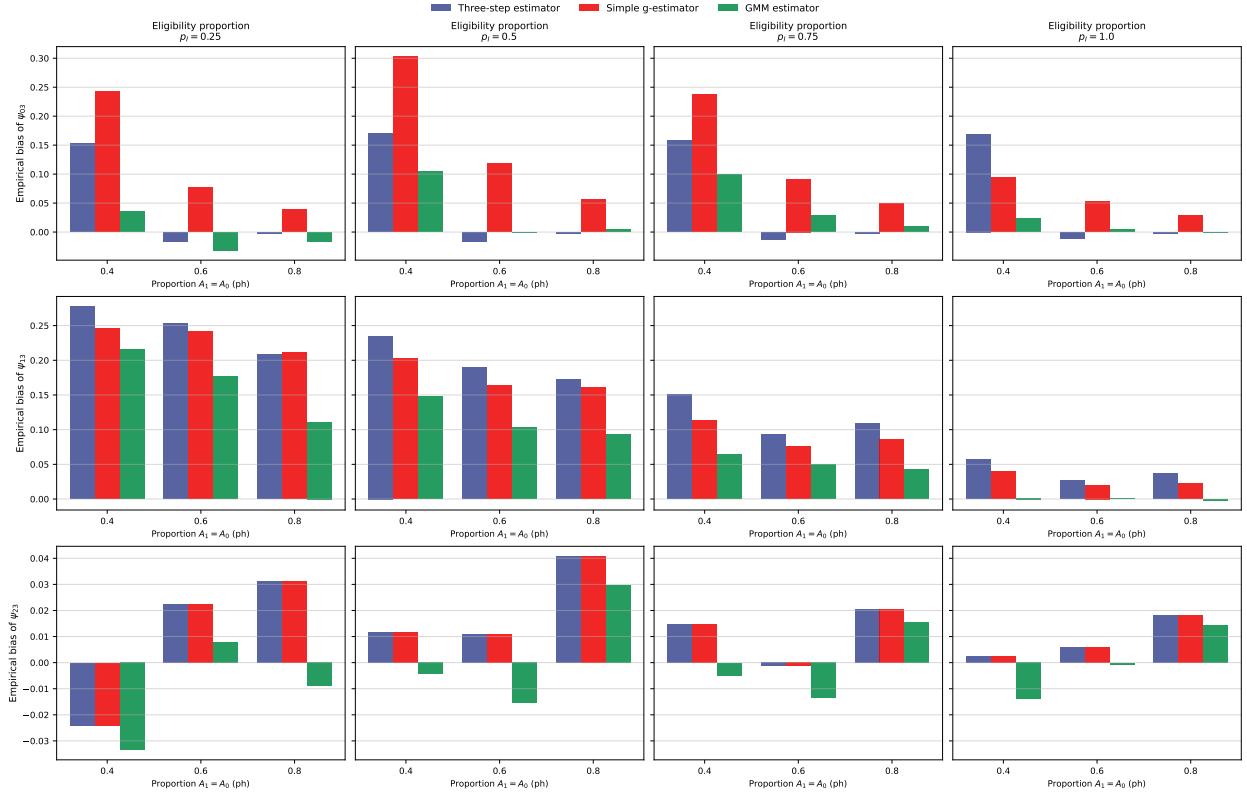


Figure 3: S2 (binary DGP, $\psi = 1$) — Empirical bias of $\hat{\psi}_{03}$, $\hat{\psi}_{13}$, and $\hat{\psi}_{23}$ (rows) as a function of p_h , stratified by p_I (columns). True values $\psi_{03} = \psi_{13} = \psi_{23} = 1$.

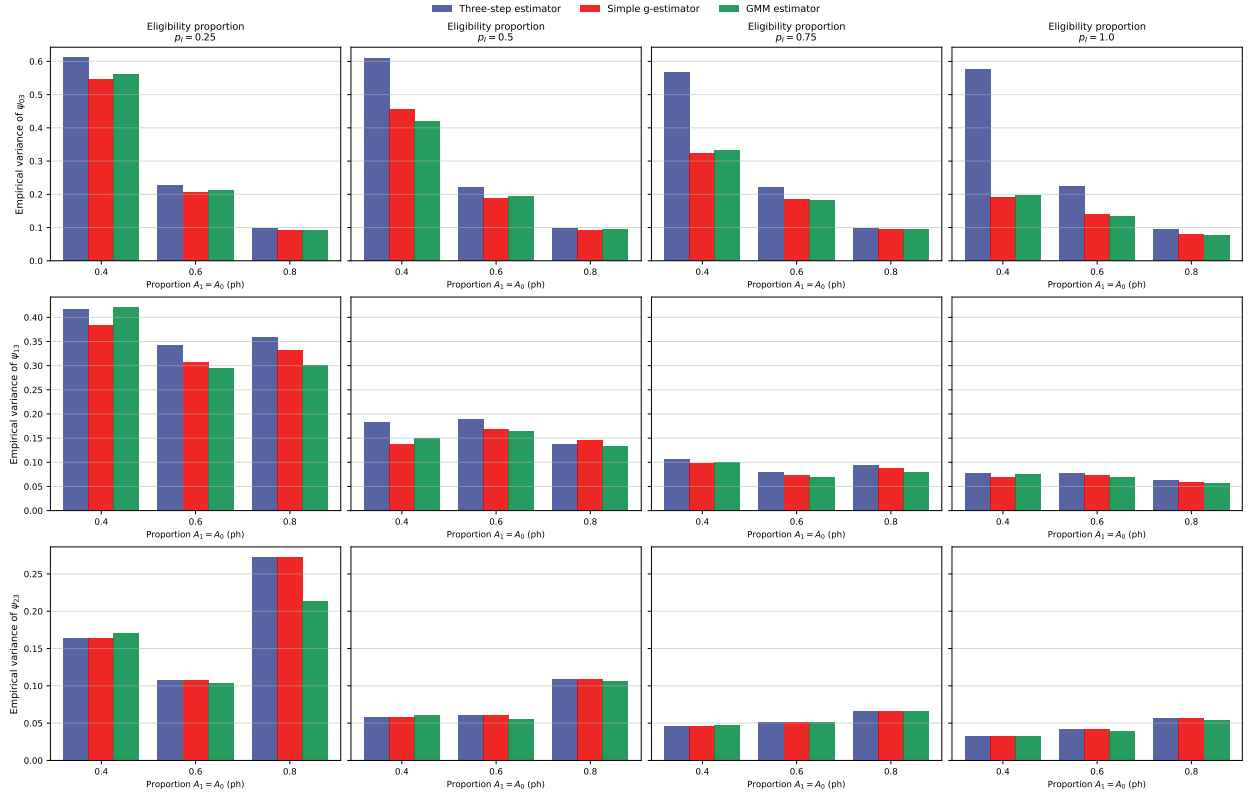


Figure 4: S2 (binary DGP, $\psi = 1$) — Empirical variance of $\hat{\psi}_{03}$, $\hat{\psi}_{13}$, and $\hat{\psi}_{23}$ (rows) as a function of p_h , stratified by p_I (columns).

Table 8: S4 ($\sigma = 10$) — Empirical bias and variance for $\hat{\psi}_{03}$. True value: $\psi_{03} = 1$.

p_h	p_I	Three-step		Simple g-est.		GMM	
		Bias	Var	Bias	Var	Bias	Var
0.4	0.25	-0.5996	22.67	-0.6513	18.65	-0.6628	18.23
	0.50	-0.6011	22.59	-0.3791	16.17	-0.3445	15.75
	0.75	-0.5907	22.70	-0.2159	19.33	-0.3146	17.28
	1.00	-0.6016	22.66	-0.1857	16.24	-0.2043	13.64
0.6	0.25	0.0431	5.353	-0.1727	5.779	-0.0709	5.225
	0.50	0.0444	5.345	-0.2589	6.758	-0.1207	5.272
	0.75	0.0424	5.340	-0.1095	5.466	-0.0142	4.166
	1.00	0.0448	5.349	-0.2782	6.126	-0.0973	4.219
0.8	0.25	0.2160	3.012	0.0533	3.270	0.1183	3.059
	0.50	0.2166	3.017	-0.0012	3.611	0.1026	3.224
	0.75	0.2156	3.012	0.0395	3.336	0.1329	2.963
	1.00	0.2144	3.011	-0.1251	3.510	0.0352	2.899

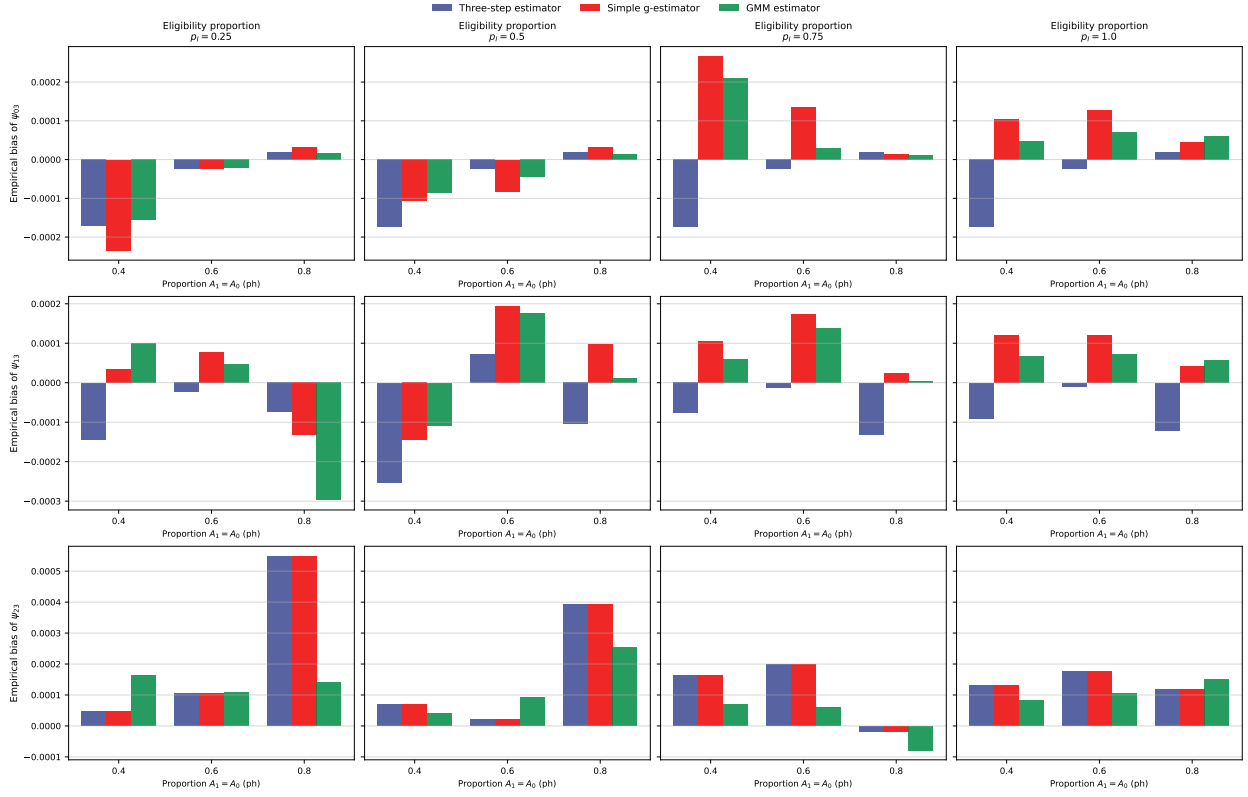


Figure 5: S3 (continuous DGP, $\sigma = 0$, $\psi = 1$) — Empirical bias of $\hat{\psi}_{03}$, $\hat{\psi}_{13}$, and $\hat{\psi}_{23}$ (rows) as a function of p_h , stratified by p_I (columns). All biases are negligibly small ($< 5 \times 10^{-4}$), confirming perfect identification in the deterministic case.

Table 9: S4 ($\sigma = 10$) — Empirical bias and variance for $\hat{\psi}_{13}$. True value: $\psi_{13} = 1$.

p_h	p_I	Three-step		Simple g-est.		GMM	
		Bias	Var	Bias	Var	Bias	Var
0.4	0.25	-0.3837	21.85	-0.2658	20.35	-0.3328	18.42
	0.50	-0.1468	11.32	-0.0535	12.31	-0.0320	11.60
	0.75	0.0825	6.975	0.3096	6.752	0.2619	6.276
	1.00	-0.0355	5.028	-0.0354	5.877	-0.0049	5.186
0.6	0.25	0.5737	16.83	0.5711	15.32	0.4188	15.39
	0.50	0.2601	7.104	0.2034	6.145	0.1475	5.631
	0.75	-0.3680	4.896	-0.1316	4.163	-0.2394	3.737
	1.00	-0.1374	3.428	-0.0665	2.947	-0.0832	2.864
0.8	0.25	0.3993	25.31	0.5403	24.42	0.2882	18.74
	0.50	0.0845	9.281	0.2481	9.398	0.0360	7.150
	0.75	-0.1501	7.008	0.0322	6.429	-0.1155	5.791
	1.00	-0.0200	4.976	0.1700	4.548	-0.0873	4.152

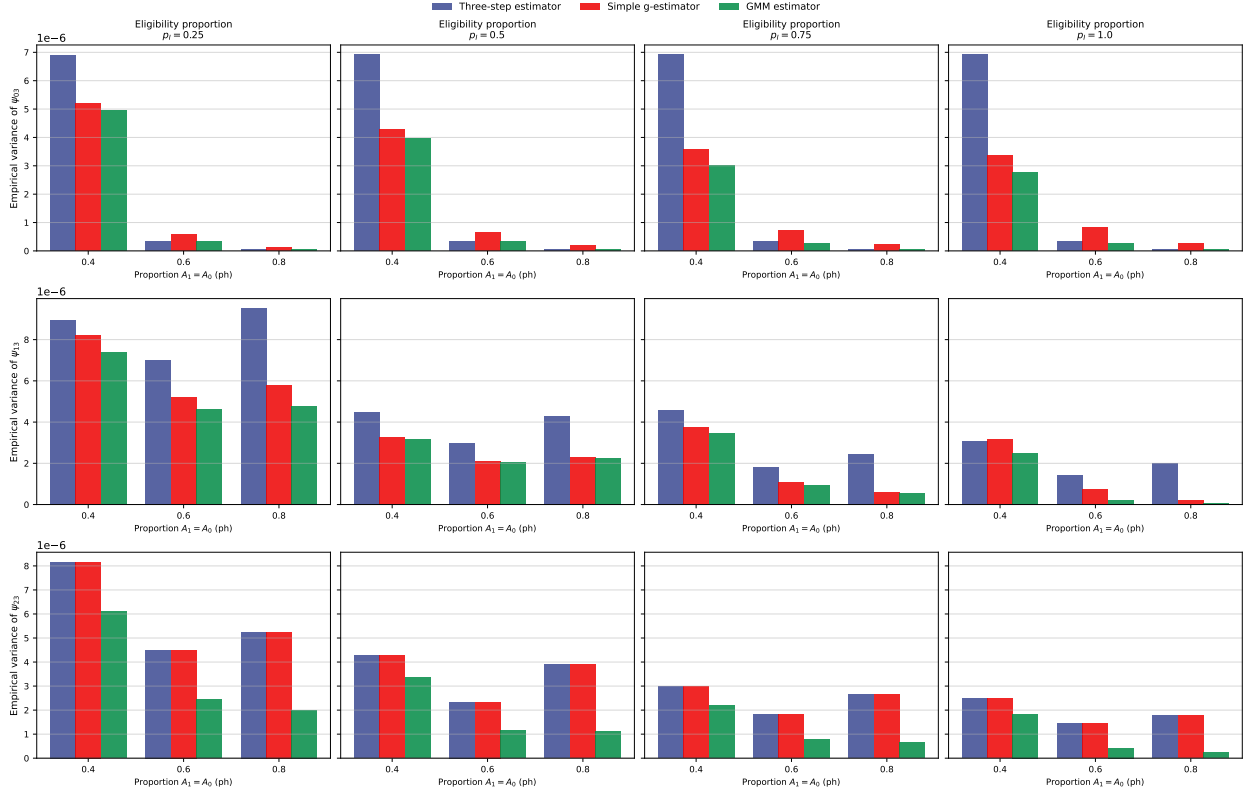


Figure 6: S3 (continuous DGP, $\sigma = 0$, $\psi = 1$) — Empirical variance of $\hat{\psi}_{03}$, $\hat{\psi}_{13}$, and $\hat{\psi}_{23}$ (rows) as a function of p_h , stratified by p_I (columns). Variances are effectively zero throughout.

Table 10: S4 ($\sigma = 10$) — Empirical bias and variance for $\hat{\psi}_{23}$. True value: $\psi_{23} = 1$. TS and SG are identical for $\hat{\psi}_{23}$ by construction.

p_h	p_I	Three-step		Simple g-est.		GMM	
		Bias	Var	Bias	Var	Bias	Var
0.4	0.25	-0.2702	22.98	-0.2702	22.98	-0.3021	20.16
	0.50	-0.1779	11.43	-0.1779	11.43	-0.1924	11.30
	0.75	0.0480	6.578	0.0480	6.578	-0.0371	6.056
	1.00	-0.0356	6.109	-0.0356	6.109	-0.0588	5.322
0.6	0.25	-0.3833	24.64	-0.3833	24.64	-0.3055	23.08
	0.50	-0.2833	10.61	-0.2833	10.61	-0.2416	9.770
	0.75	-0.0291	4.912	-0.0291	4.912	0.0500	4.590
	1.00	-0.1064	4.452	-0.1064	4.452	-0.0174	4.128
0.8	0.25	0.0247	23.49	0.0247	23.49	0.0084	22.68
	0.50	0.3811	10.68	0.3811	10.68	0.4107	10.26
	0.75	0.2574	7.587	0.2574	7.587	0.3708	6.705
	1.00	0.1806	5.616	0.1806	5.616	0.2550	4.602

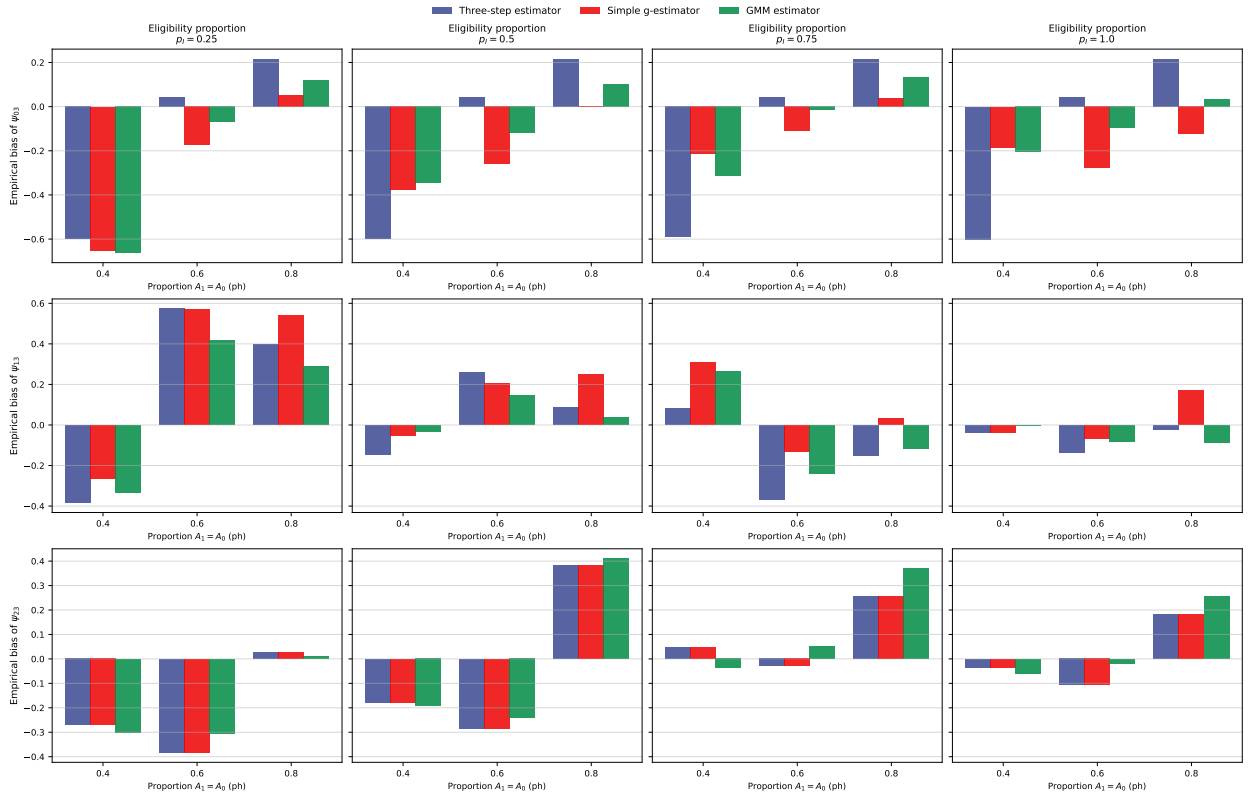


Figure 7: S4 (continuous DGP, $\sigma = 10$, $\psi = 1$) — Empirical bias of $\hat{\psi}_{03}$, $\hat{\psi}_{13}$, and $\hat{\psi}_{23}$ (rows) as a function of p_h , stratified by p_I (columns).

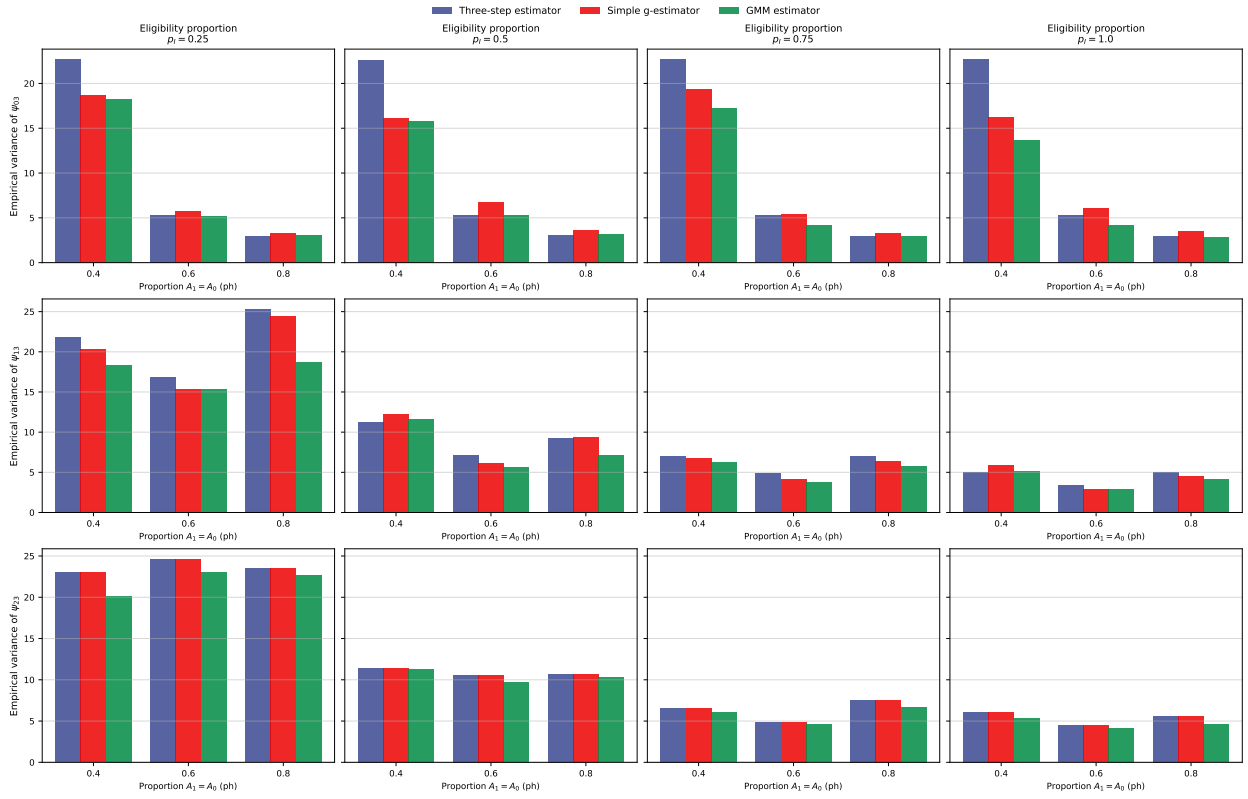


Figure 8: S4 (continuous DGP, $\sigma = 10$, $\psi = 1$) — Empirical variance of $\hat{\psi}_{03}$, $\hat{\psi}_{13}$, and $\hat{\psi}_{23}$ (rows) as a function of p_h , stratified by p_I (columns).

5.4 Scenario S4: Continuous Outcome, $\sigma = 10$ ($\psi = 1$, $n = 4000$)

5.5 Scenario S5: Continuous Outcome, $\sigma = 20$ ($\psi = 1$, $n = 4000$)

Table 11: S5 ($\sigma = 20$) — Empirical bias and variance for $\hat{\psi}_{03}$. True value: $\psi_{03} = 1$.

p_h	p_I	Three-step		Simple g-est.		GMM	
		Bias	Var	Bias	Var	Bias	Var
0.4	0.25	-1.2176	81.41	-1.2424	64.29	-1.2551	65.07
	0.50	-1.2673	82.26	-0.7245	58.17	-0.6867	57.51
	0.75	-1.1856	81.40	-0.5568	68.26	-0.8369	60.91
	1.00	-1.1994	81.60	-0.4124	59.48	-0.5286	49.79
0.6	0.25	0.1274	19.10	-0.2915	21.02	-0.0672	18.83
	0.50	0.1263	19.17	-0.4889	25.30	-0.2012	19.15
	0.75	0.1421	19.16	-0.1471	20.81	0.0330	15.16
	1.00	0.1223	19.16	-0.5098	23.53	-0.1249	15.19
0.8	0.25	0.4566	10.95	0.1357	11.89	0.2711	11.11
	0.50	0.4573	10.94	0.0197	13.32	0.2374	11.74
	0.75	0.4597	10.98	0.1117	12.37	0.2930	10.88
	1.00	0.4528	10.90	-0.2194	13.03	0.1018	10.63

6 Summary of Key Findings

Effect of eligibility proportion (p_I). Across all scenarios, increasing p_I (more eligible subjects) reduces both bias and Monte Carlo variance for $\hat{\psi}_{13}$ and $\hat{\psi}_{23}$, since these parameters are identified exclusively from eligible individuals ($I_t = 1$). The estimating equations for ψ_{03} involve all subjects, so they are less sensitive to p_I .

Effect of treatment-hold probability (p_h). The relationship between p_h and variance is non-monotone. At $p_h = 0.4$ (frequent switching), variance of $\hat{\psi}_{03}$ is large because treatment history is diverse but the baseline covariate L_0 is only weakly predictive under high switching. At $p_h = 0.8$ (high persistence), the treatment contrast $A_t - A_{t-1}$ is near zero for most subjects, reducing identification of the blip parameters and inflating variance of $\hat{\psi}_{13}$ and $\hat{\psi}_{23}$. Intermediate $p_h = 0.6$ generally yields the best variance profile.

Table 12: S5 ($\sigma = 20$) — Empirical bias and variance for $\hat{\psi}_{13}$. True value: $\psi_{13} = 1$.

p_h	p_I	Three-step		Simple g-est.		GMM	
		Bias	Var	Bias	Var	Bias	Var
0.4	0.25	-0.9476	76.13	-0.6035	75.40	-0.7694	66.26
	0.50	-0.4827	39.26	-0.2183	45.16	-0.2317	40.68
	0.75	0.1019	26.42	0.5321	25.58	0.3501	23.54
	1.00	-0.1481	18.69	-0.1006	21.90	-0.1139	18.91
0.6	0.25	1.0316	61.25	1.0058	55.79	0.7272	56.32
	0.50	0.4784	26.11	0.3827	22.56	0.2721	20.52
	0.75	-0.6959	18.46	-0.2572	15.09	-0.4633	13.74
	1.00	-0.2833	12.98	-0.1290	10.68	-0.1666	10.27
0.8	0.25	0.8216	92.57	1.1173	89.51	0.6100	68.70
	0.50	0.2059	34.41	0.5399	34.73	0.1085	26.31
	0.75	-0.2120	26.14	0.1246	24.21	-0.1724	21.92
	1.00	0.0235	18.57	0.3810	17.01	-0.1377	15.63

Table 13: S5 ($\sigma = 20$) — Empirical bias and variance for $\hat{\psi}_{23}$. True value: $\psi_{23} = 1$. TS and SG are identical for $\hat{\psi}_{23}$ by construction.

p_h	p_I	Three-step		Simple g-est.		GMM	
		Bias	Var	Bias	Var	Bias	Var
0.4	0.25	-0.6417	88.02	-0.6417	88.02	-0.6751	77.18
	0.50	-0.3208	44.05	-0.3208	44.05	-0.3807	43.84
	0.75	0.0853	24.52	0.0853	24.52	-0.1295	22.60
	1.00	-0.1009	23.18	-0.1009	23.18	-0.1932	20.14
0.6	0.25	-0.7927	92.20	-0.7927	92.20	-0.6192	86.01
	0.50	-0.5401	39.34	-0.5401	39.34	-0.4626	36.21
	0.75	-0.0271	18.08	-0.0271	18.08	0.1131	16.77
	1.00	-0.1969	16.47	-0.1969	16.47	-0.0098	15.09
0.8	0.25	0.0086	89.17	0.0086	89.17	-0.0293	86.42
	0.50	0.7390	40.01	0.7390	40.01	0.8088	38.37
	0.75	0.5356	28.43	0.5356	28.43	0.7438	25.05
	1.00	0.3669	20.86	0.3669	20.86	0.4911	17.01

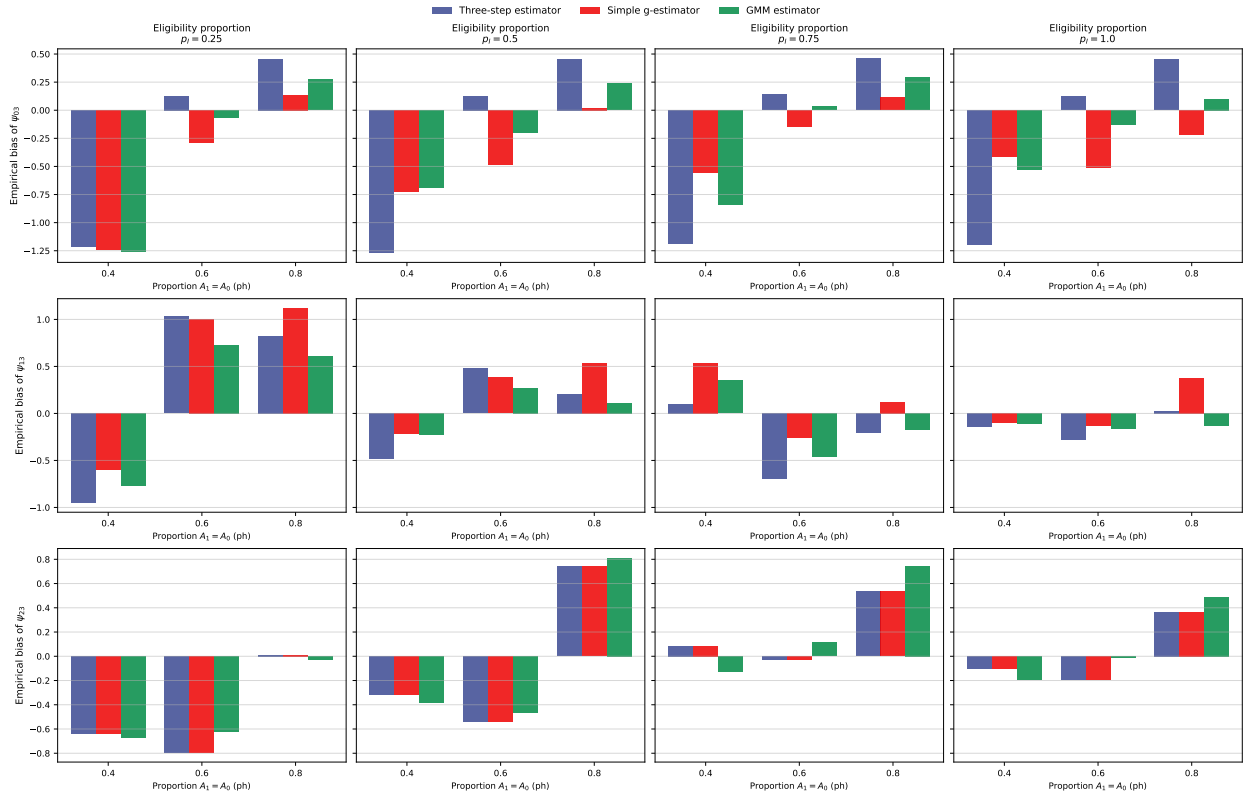


Figure 9: S5 (continuous DGP, $\sigma = 20$, $\psi = 1$) — Empirical bias of $\hat{\psi}_{03}$, $\hat{\psi}_{13}$, and $\hat{\psi}_{23}$ (rows) as a function of p_h , stratified by p_I (columns). Biases are substantially larger than in S4 ($\sigma = 10$), scaling approximately proportionally with σ .

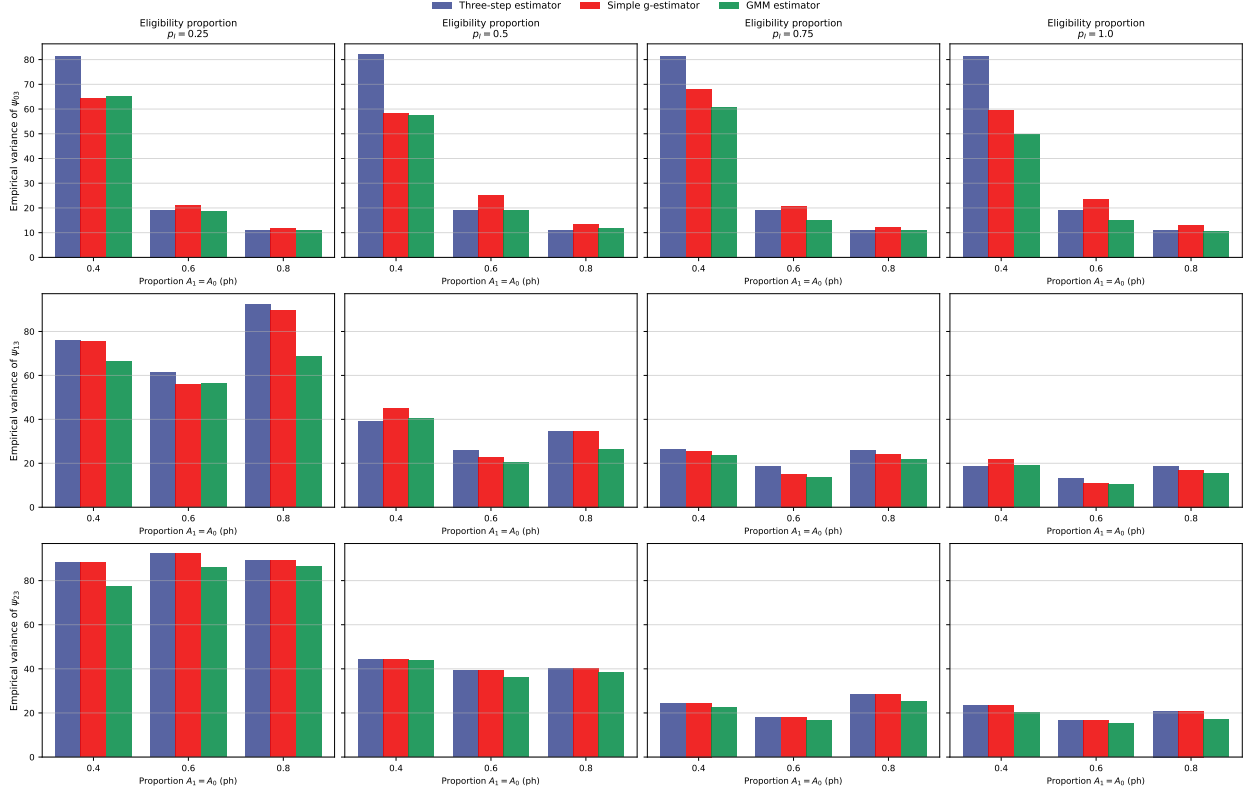


Figure 10: S5 (continuous DGP, $\sigma = 20$, $\psi = 1$) — Empirical variance of $\hat{\psi}_{03}$, $\hat{\psi}_{13}$, and $\hat{\psi}_{23}$ (rows) as a function of p_h , stratified by p_I (columns). Variances are roughly four times larger than in S4, consistent with $O(\sigma^2)$ variance scaling.